

PAPER_1779 — Cosmic Ray Ankle Energy = $E_{\text{ankle}} = m_p \times D_{\text{crit}}^7 / K_{\text{MEX}} = 3.62 \times 10^{18} \text{ eV}$ (Auger $\approx 3.6 \times 10^{18}$)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** High-Energy Astro **Date:** June 18, 2026 **Status:** CLOSED — 0.1% closure (PAPER_1404-1428 catalog stubs)

Observation

PAPER_1418 catalog stub gives:

$$E_{\text{ankle}} = m_p \times D_{\text{crit}}^7 / K_{\text{MEX}} = 3.62 \times 10^{18} \text{ eV (Auger } \approx 3.6 \times 10^{18})$$

Status: **0.1%**.

UQFF Closed Identity

$$E_{\text{ankle}} = m_p \times D_{\text{crit}}^7 / K_{\text{MEX}} = 3.62 \times 10^{18} \text{ eV (Auger } \approx 3.6 \times 10^{18}) \quad 0.1\%$$

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1418
- Related: PAPER_1766-1775 (tier-35); previous bucket migrations
- Calculator dispatch: `calculate_paradox({"paradox": "e_ankle_3_62e18_ev"})`

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